Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Program: B.Tech Vth Semester
Paper Name: Probability Theory and Stochastic Processes
Course Code: TEC-503
Time: 1-1/2 Hour

MM:50

Note:

(i) Answer **all the questions** by choosing **any one of the sub questions.**
(ii) Each question carries 10 marks.

Q.1 **(10 Marks)**

a) Discuss the following Laws/Theorem of probability with illustrative examples: (CO 1)
(i).Addition law
(ii).Multiplication law
(iii).Baye's theorem

OR

b) A deck of 52 standard playing cards is shuffled, and one card is drawn at random. Find the probability that the card drawn is either a queen or a diamond.

Q.2 **(10 Marks)**

a) What is the difference between independent and mutually exclusive events? Discuss with suitable examples. (CO 1)

OR

b) A company has three machines M1, M2, and M3 that produce 20%, 30%, and 50% of its total products, respectively.Machine M1 produces 8% defective items, M2 produces 4% defective items, and M3 produces 6% defective items.
If a product is chosen at random and is found to be defective, what is the probability that it was produced by Machine M1?

Q.3 **(10 Marks)**

a) Discuss Poisson distribution. Consider a network where packets arrive according to a Poisson distribution with a mean arrival rate of 3 packets per second. What is the probability that exactly 4 packets arrive in the first second? (CO2)

OR

b) What do you mean by a Probability mass function, and Cumulative distribution function of a discrete random variable? Explain with an example.

Q.4 **(10 Marks)**

a) (i).What do you mean by the Binomial distribution? (CO 2)
(ii). A fair coin is flipped 5 times. What is the probability of getting exactly 3 heads?

OR

b) Derive the expression for mean of a Binomial distribution.

Q.5

a) Consider a discrete random variable Y = {1,3,5,7,9} with corresponding probabilities $P_Y(y)$ = {0.4, 0.2, 0.2, 0.1, 0.1}. Determine mean, variance, and CDF of Y. (CO 2)

OR

b) Write short notes on:
(i).Sample space
(ii). Random variable
(iii).Expected value